NET732

Week 7:Mideterm Practical

Practical

1. Checking system information

I started by checking my Ubuntu system information using:

date

hostname

ip addr

```
maso-prince@maso-prince-VMware-Virtual-Platform:~$ mkdir -p ~/CyberOps/Week7
maso-prince@maso-prince-VMware-Virtual-Platform:~$ cd ~/CyberOps/Week7
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week7$ date
Fri Oct 30 06:34:18 PM SAST 2026
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week7$ hostname
maso-prince-VMware-Virtual-Platform
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week7$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 1000
    link/ether 00:0c:29:4a:af:9c brd ff:ff:ff:ff:ff:ff
    inet 192.168.134.128/24 brd 192.168.134.255 scope global dynamic noprefixroute ens33
       valid_lft 1356sec preferred_lft 1356sec
    inet6 fe80::20c:29ff:fe4a:af9c/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week7$
```

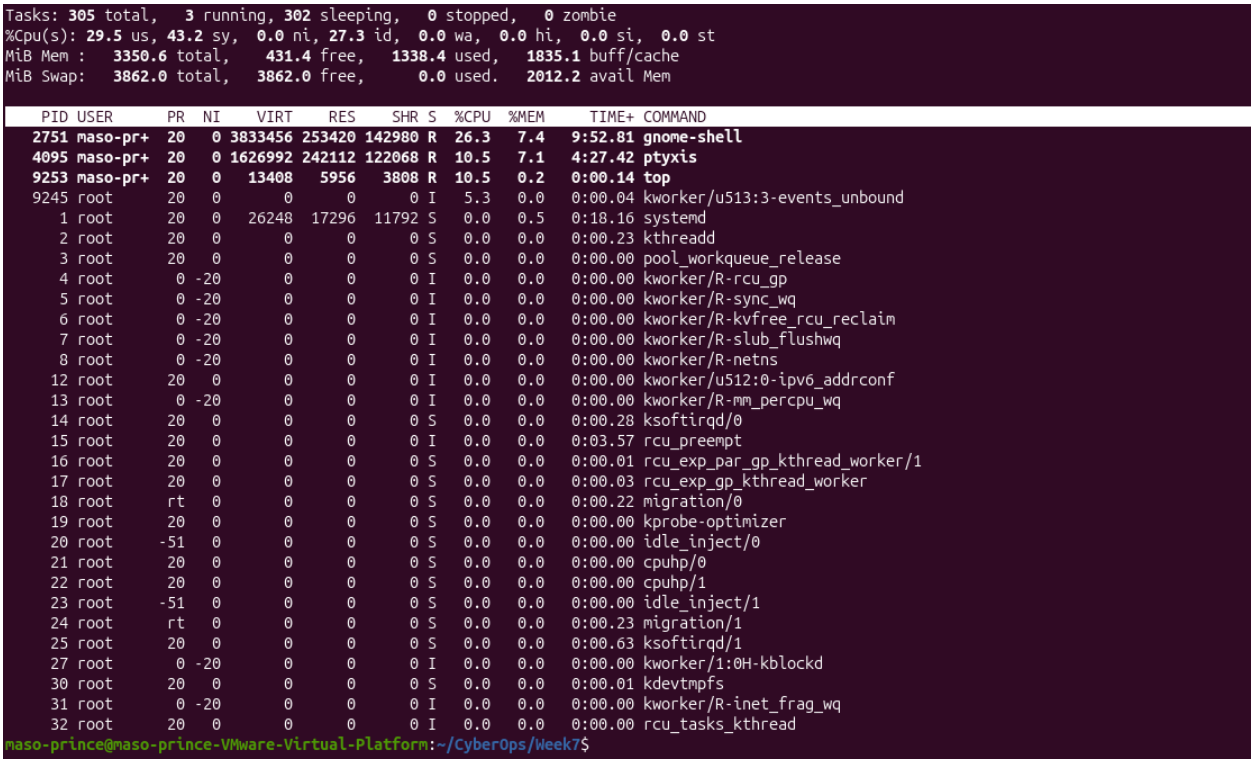
These commands helped me identify my computer name, date and network configuration.

2. Investigating running processes

I used the following commands:

ps aux

top



I used these commands to view running processes and check how my computer was using its resources.

3. Checking network connections

I used:

ss -tulnp

ss -tunap

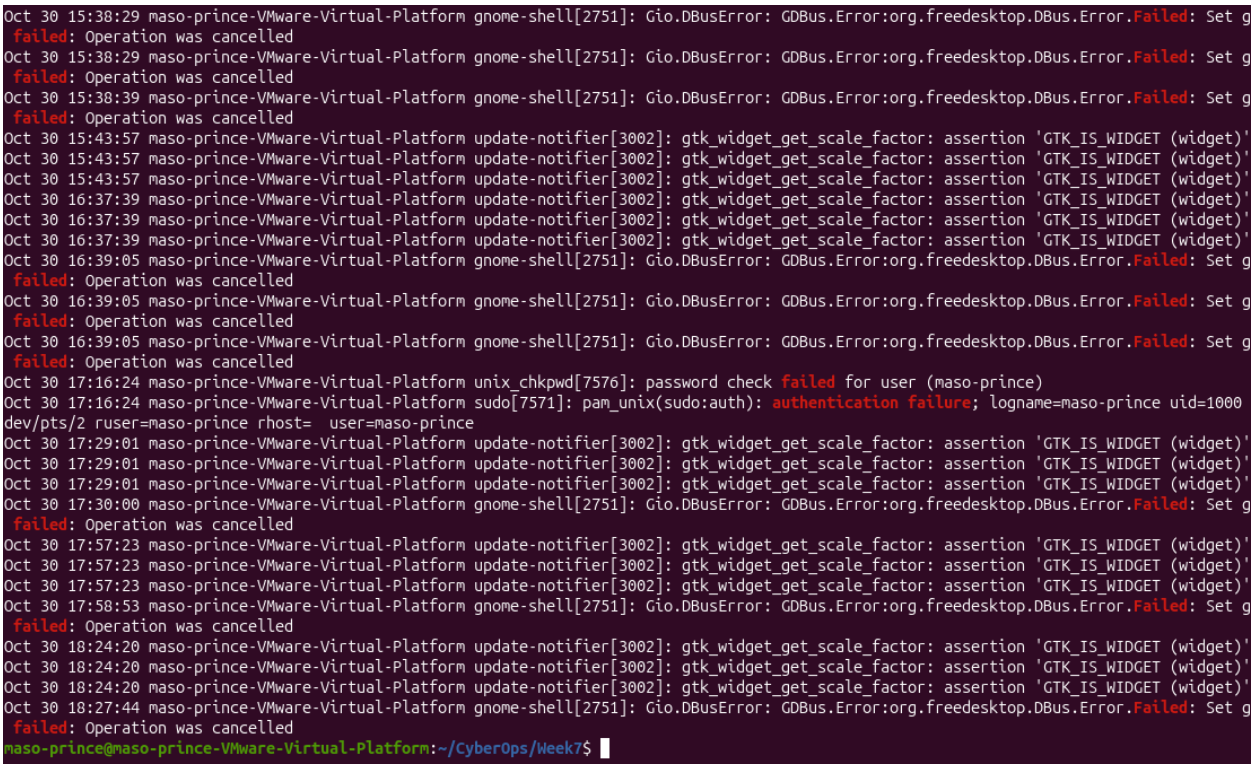
```
Netid    State     Recv-Q    Send-Q                          Local Address:Port          Peer Address:Port    Process
udp      UNCONN    0         0                                     0.0.0.0:49937              0.0.0.0:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                                     0.0.0.0:5353               0.0.0.0:*
udp      UNCONN    0         0                                192.168.134.128:3702            0.0.0.0:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                                239.255.255.250:3702            0.0.0.0:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                                    127.0.0.54:53               0.0.0.0:*
udp      UNCONN    0         0                                 127.0.0.53%lo:53               0.0.0.0:*
udp      UNCONN    0         0                                    127.0.0.1:323               0.0.0.0:*
udp      UNCONN    0         0                                              *:55981                  *:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                                           [::]:5353                [::]:*
udp      UNCONN    0         0                 [fe80::20c:29ff:fe4a:af9c]%ens33:3702              [::]:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                              [ff02::c]%ens33:3702                  [::]:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                                           [::1]:323                [::]:*
tcp      LISTEN    0         4096                                 127.0.0.54:53               0.0.0.0:*
tcp      LISTEN    0         4096                              127.0.0.53%lo:53               0.0.0.0:*
tcp      LISTEN    0         4096                                 127.0.0.1:631               0.0.0.0:*
tcp      LISTEN    0         4096                                      [::1]:631                [::]:*
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week7$ ss -tunap
Netid    State     Recv-Q    Send-Q                          Local Address:Port          Peer Address:Port    Process
udp      UNCONN    0         0                                     0.0.0.0:49937              0.0.0.0:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                                     0.0.0.0:5353               0.0.0.0:*
udp      UNCONN    0         0                                192.168.134.128:3702            0.0.0.0:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                                239.255.255.250:3702            0.0.0.0:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                                    127.0.0.54:53               0.0.0.0:*
udp      UNCONN    0         0                                 127.0.0.53%lo:53               0.0.0.0:*
udp      ESTAB     0         0                         192.168.134.128%ens33:68          192.168.134.254:67
udp      UNCONN    0         0                                    127.0.0.1:323               0.0.0.0:*
udp      UNCONN    0         0                                              *:55981                  *:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                                           [::]:5353                [::]:*
udp      UNCONN    0         0                 [fe80::20c:29ff:fe4a:af9c]%ens33:3702              [::]:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                              [ff02::c]%ens33:3702                  [::]:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                                           [::1]:323                [::]:*
tcp      LISTEN    0         4096                                 127.0.0.54:53               0.0.0.0:*
tcp      LISTEN    0         4096                              127.0.0.53%lo:53               0.0.0.0:*
tcp      LISTEN    0         4096                                 127.0.0.1:631               0.0.0.0:*
tcp      LISTEN    0         4096                                      [::1]:631                [::]:*
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week7$
```

These commands helped me identify open ports, listening services and active network connections.

4. Investigating authentication logs

I used the following command:

sudo journalctl | grep -iE "failed|invalid user|authentication failure"



I used it to search for failed login attempts and unusual authentication activity.

5. Capturing network traffic

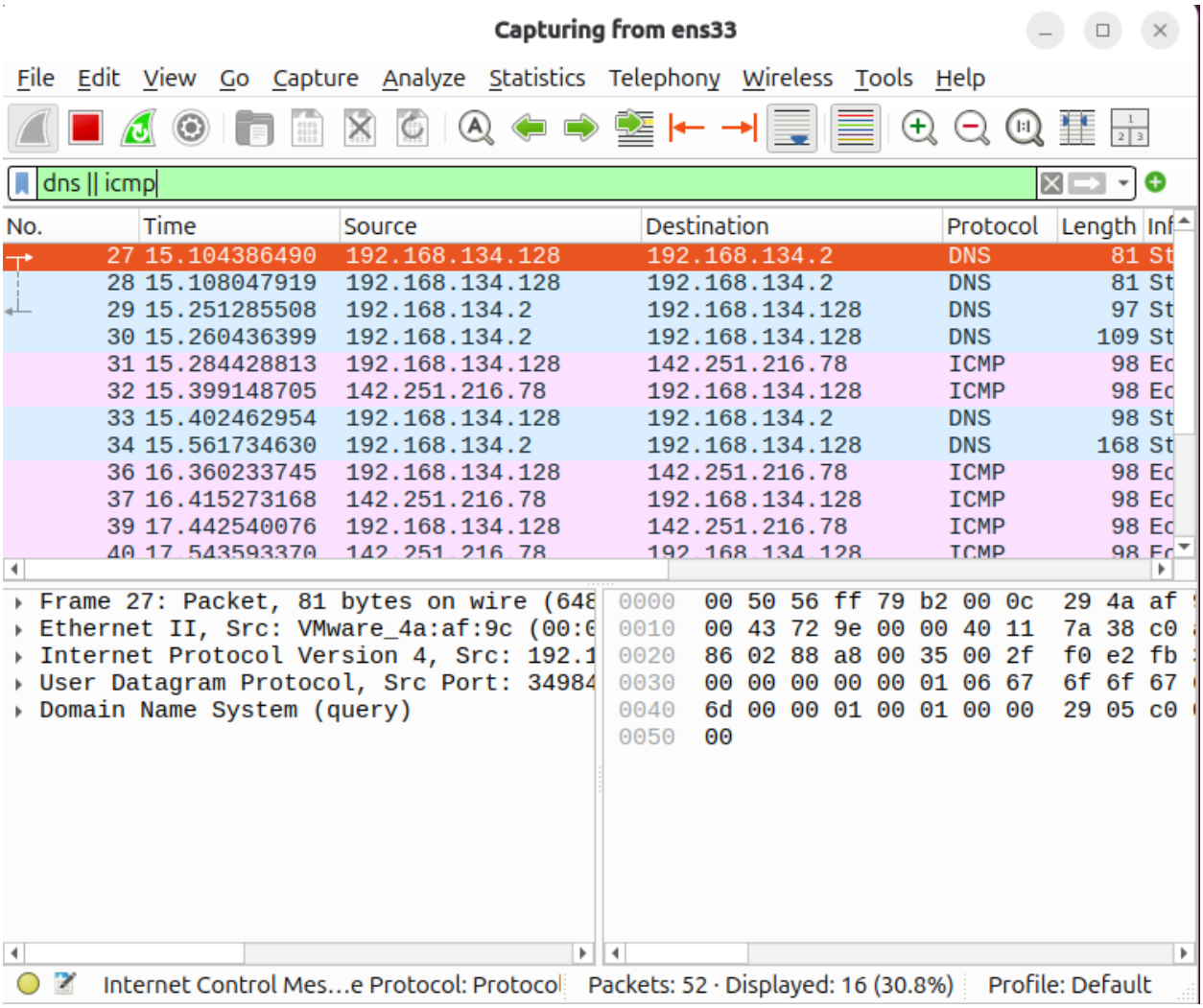
I opened Wireshark, selected my active network interface and started capturing packets.

I then generated network traffic using:

ping -c 5 google.com

I used the Wireshark filter:

dns || icmp



This allowed me to view DNS and ICMP traffic. I saved my capture as Week7_Incident_Capture.pcapng.

6. Saving investigation results

I saved my investigation results using:

ps aux > processes.txt

ss -tulnp > network_connections.txt

sudo journalctl -n 50 > system_logs.txt

I also recorded my terminal activities in commands.log.

7. Analysis and conclusion

During this practical, I investigated my Ubuntu system by checking running processes, network connections and authentication logs. I also used Wireshark to capture and analyse network traffic.

I learned how different tools can help identify unusual system activities and possible security threats. This is important in SOC operations because security analysts use evidence from systems and networks to investigate incidents.

I successfully completed the investigation and saved my evidence. My next step is to improve my ability to identify suspicious activities and respond to potential security.